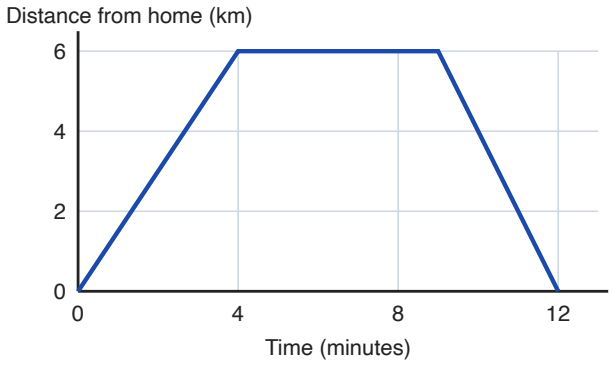
Graphs 3 — Distance–time graphs · Speed–time graphs

Calculator allowed. Squared paper for the drawing question. State units in every answer. Marks in brackets. ★ = harder question.

1 The travel graph shows Anjali's journey to a friend's house and back. [6]



She leaves at 0 min, arrives (6 km) at 4 min, stays until 9 min, and is home at 12 min.

a Calculate her outward speed in km/min.

b How long did she stay at her friend's house?

c Calculate her return speed in km/h.

2 A drone accelerates steadily from rest to 15 m/s in 25 s, flies at that speed for 35 s, then slows steadily to rest in a further 40 s. [10]

a Draw the speed–time graph of the flight.

b Use the graph to find the drone's **(i)** initial acceleration, **(ii)** final acceleration, **(iii)** average speed over the whole 100 s flight.

3 The winner of a 400 m race finished in 50.0 seconds. [9]

a Calculate his average speed in m/s.

b Change your answer to km/h.

c The sketch of his speed–time graph shows him accelerating uniformly from rest for the first 6 seconds, then running at a constant maximum speed for the rest of the race. Calculate his maximum speed, correct to 3 s.f.

d Calculate his acceleration over the first 6 seconds, with units, correct to 2 s.f.

4 A train leaves a station and accelerates uniformly for 2 minutes until it reaches 30 m/s, travels at 30 m/s for 10 minutes, then decelerates uniformly to rest in 90 seconds at the next station. Calculate the distance between the stations in kilometres. [5]

Total: 30 marks

Check: distance–time — gradient = speed, flat = stopped, down = returning · speed–time — gradient = acceleration, *area* = distance · convert minutes to seconds (or hours) before calculating · m/s → km/h: × 3.6.